3.1.3	**Bonding**	
	Nature of ionic, covalent and metallic bonds	understand that ionic bonding involves attraction between oppositely charged ions in a lattice
		know that a covalent bond involves a shared pair of electrons
		know that co-ordinate bonding is dative covalency
		understand that metallic bonding involves a lattice of positive ions surrounded by delocalised electrons
	Bond polarity	understand that electronegativity is the power of an atom to withdraw electron density from a covalent bond
		understand that the electron distribution in a covalent bond may not be symmetrical
		know that covalent bonds between different elements will be polar to different extents
	Forces acting between molecules	understand qualitatively how molecules may interact by permanent dipole–dipole, induced dipole–dipole (van der Waals') forces and hydrogen bonding
		understand the importance of hydrogen bonding in determining the boiling points of compounds and the structures of some solids (e.g. ice)

	States of matter	be able to explain the energy changes associated with changes of state
		recognise the four types of crystal: ionic, metallic, giant covalent (macromolecular) and molecular
		know the structures of the following crystals: sodium chloride, magnesium, diamond, graphite, iodine and ice
		be able to relate the physical properties of materials to the type of structure and bonding present
	Shapes of simple molecules and ions	understand the concept of bonding and lone (non bonding) pairs of electrons as charge clouds.
		be able, in terms of electron pair repulsion, to predict the shapes of, and bond angles in, simple molecules and ions, limited to 2, 3, 4, 5 and 6 co-ordination
		know that lone pair/lone pair repulsion is greater than lone pair/bonding pair repulsion, which is greater than bonding pair/bonding pair repulsion, and understand the resulting effect on bond angles